Differentiation from First Principles

Exam code: 9MA0

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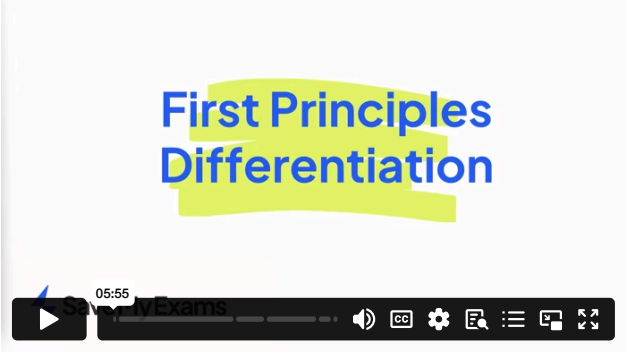

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Differentiation from first principles

What is the derivative or gradient function?

- For a curve **y = f(x)** there is an associated function called the **derivative** or **gradient function**
- The derivative of f(x) is written as f'(x)
- The derivative is a formula that can be used to find the gradient of y = f(x) at any point, by substituting the x coordinate of the point into the formula
- The process of finding the derivative of a function is called differentiation
- We differentiate a function to find its derivative

What is differentiation from first principles?

- Differentiation from **first principles** uses the **definition** of the **derivative** of a function ***f(x)***
- The definition is

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

undefined

- Instead we consider what happens as ***h*** gets closer and closer to zero

- Differentiation **from first principles** means using that definition to show what the **derivative** of a function is

How do I differentiate from first principles?

- **STEP 1**
  Identify the function ***f(x)*** and substitute this into the first principles formula
  - e.g. Show, from first principles, that the derivative of $3x^2$ is $6x$
  - $f(x) = 3x^2$ so
    $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{3(x+h)^2 - 3x^2}{h}$

- **STEP 2**
  Expand ***f(x+h)*** in the numerator
  - $f'(x) = \lim_{h \to 0} \frac{3(x^2 + 2hx + h^2) - 3x^2}{h}$
  - $f'(x) = \lim_{h \to 0} \frac{3x^2 + 6hx + 3h^2 - 3x^2}{h}$

- **STEP 3**
  Simplify the numerator, factorise and cancel h with the denominator
  - $f'(x) = \lim_{h \to 0} \frac{h(6x + 3h)}{h}$

- **STEP 4**
  Evaluate the remaining expression as ***h*** *tends to zero*
  - $f'(x) = \lim_{h \to 0} (6x + 3h) = 6x$
    $As\ h \to 0,\ (6x + 3h) \to (6x + 0) \to 6x$
  - $\therefore$ The derivative of $3x^2$ is $6x$

Examiner Tips and Tricks

- Most of the time you will not use first principles to find the derivative of a function (there are much quicker ways!). However, you can be asked on the exam to demonstrate differentiation from first principles.
- Make sure you can use first principles differentiation to find the derivatives of kx, kx^2 and kx^3 (where k is a constant).

Worked Example

the derivative of $5x^3$ is $15x^2$.

STEP 1: FOR $f(x) = 5x^3$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{5(x+h)^3 - 5x^3}{h}$$

EXPAND $(x+h)^3$

STEP 2:

$$= \lim_{h \to 0} \frac{5(x^3 + 3x^2h + 3xh^2 + h^3) - 5x^3}{h}$$

$$= \lim_{h \to 0} \frac{5x^3 + 15x^2h + 15xh^2 + 5h^3 - 5x^3}{h}$$

STEP 3:

$$= \lim_{h \to 0} \frac{15x^2h + 15xh^2 + 5h^3}{h}$$

$$= \lim_{h \to 0} 15x^2 + 15xh + 5h^2$$

TAKE THE LIMIT AS $h \to 0$

STEP 4: AS $h \to 0$

$$(15x^2 + 15xh + 5h^2) \longrightarrow (15x^2 + 15x(0) + 5(0)^2) = 15x^2$$

$\therefore$ THE DERIVATIVE OF $5x^3$ IS $15x^2$

TO GET FULL MARKS, MAKE SURE YOU KEEP WRITING $\lim_{h \to 0}$ RIGHT UP TO THE END!

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